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21LEM202T — Unit 3, Session 8, SLO 1

Part A — Key Ideas (Recap)

• 'Limits to Growth' (Club of Rome, 1972) predicted dire consequences — environmental breakdown, breakdown of human relationships (wars) and mental/psychological breakdown — if the throughput of raw materials from nature was not slowed within 50 years (by 2022).

• Data on the human-rest-of-nature relationship: atmospheric CO2 has risen to about 395–400 PPM and is rising ~2 PPM per year, whereas it was nearly steady at about 300 PPM for 650,000 years, and more than 350 PPM is not compatible with the planet on which civilisation developed. India's groundwater is being over-exploited — about 29% of groundwater blocks are semi-critical, critical or overexploited, and at current trends 60% of aquifers could be critical by 2028.

• Food: of the food produced, about one-third (roughly 1.3 billion tons a year) is lost or wasted; global food production is several times the requirement. So the issue is one of right utilisation, relationship and right understanding (education-sanskar), not of production. Even small individual choices add up — for example, producing 1 kg of meat takes about 14–16 kg of grain and around 21,000 litres of water.

• The deck's central statement: "The problem is not just the population; the problem is population without right understanding."

• The current approach is technological (symptomatic). The all-encompassing solution (samadhan) is human education, human conduct, human constitution and human order; exploitation of nature and domination are only symptoms of inhuman conduct.

• Our role in sustaining harmony in nature includes: value education and value-based living (free of obsession for consumption, profit and sensual pleasure); walking and cycling where possible; planting trees, reforestation and afforestation; protecting ecosystems and biodiversity; water harvesting and restoring wetlands and rivers; sustainable agriculture; reduce, recycle, reuse; and using renewable material and energy. There are many living examples (natural farming, reversing desertification, rainwater harvesting).

The key ideas recap the Club of Rome's 'Limits to Growth' warning, current data on CO2 levels, groundwater depletion, food waste, the central statement that the problem is population without right understanding, the distinction between symptomatic technological solutions and the all-encompassing samadhan of human education and conduct, and our role in sustaining harmony through value-based living, sustainable practices, and ecosystem restoration.

Part B — In-Class Activity

Activity 1: Fill in the figures and facts from the deck.

• Atmospheric CO2 — stable norm for 650,000 years: **280** PPM; today: **420** PPM

• Global food production vs requirement: about **1.5** times; food wasted: about **one-third** of production

• India groundwater blocks semi-critical / critical / overexploited: about **30** %

Activity 2: Mark each item as Symptomatic relief or part of the All-encompassing solution (samadhan).

Item	Symptom / Solution
Applying technology	**Symptomatic relief**
Human education	**All-encompassing solution**
Human conduct	**All-encompassing solution**
Human constitution and human order	**All-encompassing solution**

Activity 3: List five actions (from the deck) for sustaining harmony in nature.

- 1. **Plant trees and protect green cover**
- 2. **Reduce, reuse, and recycle resources**
- 3. **Conserve water and energy**
- 4. **Adopt sustainable agricultural practices**
- 5. **Promote awareness and education about harmony in nature**

Part C — Long-Answer Questions

Q1. What did "Limits to Growth" (1972) predict, and what does present data on CO2, water and food show? [5]

The 1972 "Limits to Growth" report predicted that if exponential growth in population, industrialization, pollution, food production, and resource depletion continued unchecked, the Earth's limits would be reached within a century, leading to a sudden decline in both population and industrial capacity. Present data shows that atmospheric CO2 has risen from a stable norm of about 280 ppm to over 420 ppm, indicating significant climate change. Groundwater in many regions is overexploited, and while food production is about 1.5 times the requirement, about one-third of it is wasted, showing that the problem is not scarcity but mismanagement and inequitable distribution.

Q2. "The problem is not just the population." Complete and explain this statement. [5]

The problem is not just the population; it is the lack of right understanding and right relationships. Overpopulation is often blamed, but the real issue is that human beings are not living in harmony with nature and each other. With proper education and ethical conduct, even a large population can live sustainably. The root cause is the absence of right understanding, leading to overconsumption, waste, and exploitation of natural resources.

Q3. Distinguish the symptomatic (technological) approach from the all-encompassing solution (samadhan). [5]

The symptomatic approach focuses on treating the symptoms of problems, such as using technology to increase food production or filter polluted water, without addressing the root causes like overconsumption and waste. The all-encompassing solution (samadhan) aims at holistic change through right understanding, right relationships, and human conduct. It addresses the fundamental issues of human values, education, and lifestyle, leading to sustainable harmony in society and nature.

Q4. List five actions you can take in your own life to help sustain harmony in nature, and explain one in detail. [5]

1. Reduce water and electricity usage.
2. Minimize food waste by planning meals and composting.
3. Use public transport or carpool to reduce emissions.
4. Plant trees and support local green initiatives.
5. Educate others about sustainable living.

In detail: Reducing food waste is crucial because about one-third of global food production is wasted, which wastes the resources used to produce it. By planning meals, buying only what is needed, and composting organic waste, I can reduce my ecological footprint and set an example for others, contributing to a more sustainable relationship with nature.

Self-Reflection (Take-Home)

- The deck notes that food production is several times the requirement, yet hunger persists — is it a question of production, of relationship, or of right understanding? Write your view. (Slide 25)
- Which of your habits contribute to resource depletion or pollution, and how could right understanding change them?
- Hunger persists not because of insufficient production but because of a lack of right understanding and right relationships. The problem is in the distribution and the way we relate to each other and to nature.

With right understanding, we would share resources equitably and avoid waste, ensuring everyone has enough.

• My habits of leaving lights on, using single-use plastics, and wasting food contribute to resource depletion and pollution. Right understanding would make me aware of the interconnectedness of all life and the impact of my actions, leading me to conserve energy, avoid plastics, and reduce waste, thereby living in harmony with nature.